

35. Define Main Memory.

Ans:

Main memory is where Programs & data are kept when the Processor is actively using them.

36. Write the Properties of memory.

Ans: The Properties of memory are

a. Private

b. Large

c. Fast

d. Non-volatile

e. Cheap.

37. What do you mean by Volatile & Non-volatile.

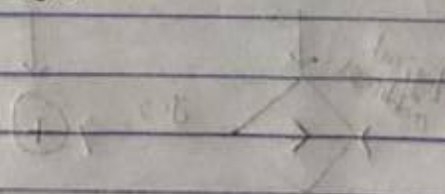
Ans:

Volatile memory is temporary memory.

Example: RAM.

Non-volatile memory keeps its contents even if the power is lost.

Example: Hard disk.



38. Write the drawbacks of memory relocation.

Ans:

The drawbacks are

- If users Programs can address every byte of memory, they can easily crash the operating system.

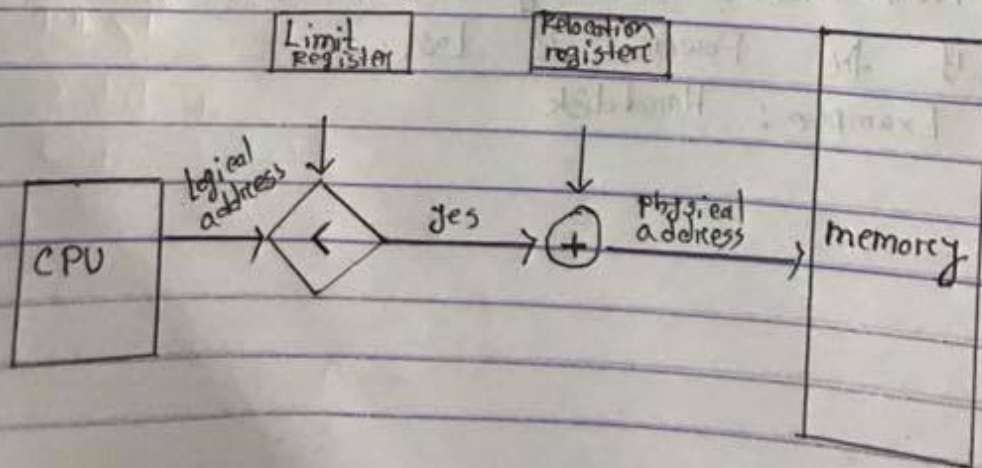
- It is difficult to have run multiple Programs.

39. How static relocation works?

Ans:-

It loads the first instruction of Program at address x & add x to every subsequent address during loading.

Base & Limit registers
Load করে পূর্ণা যাত্র



40. What is the duty of Memory Management Unit

Ans:-

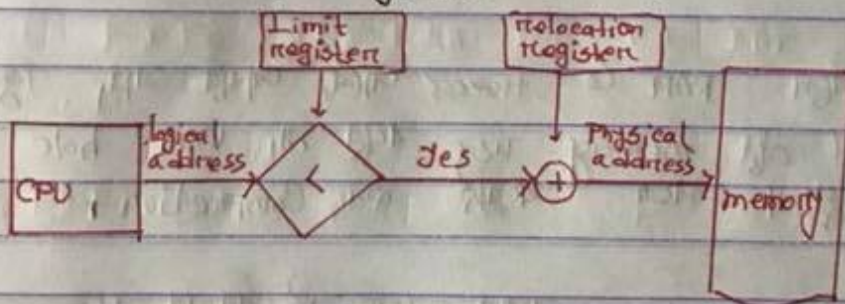
- Decide which Process will go where.
- Space allocation.
- Delete Space for a Process.

41. How a Process can crash.

Ans:

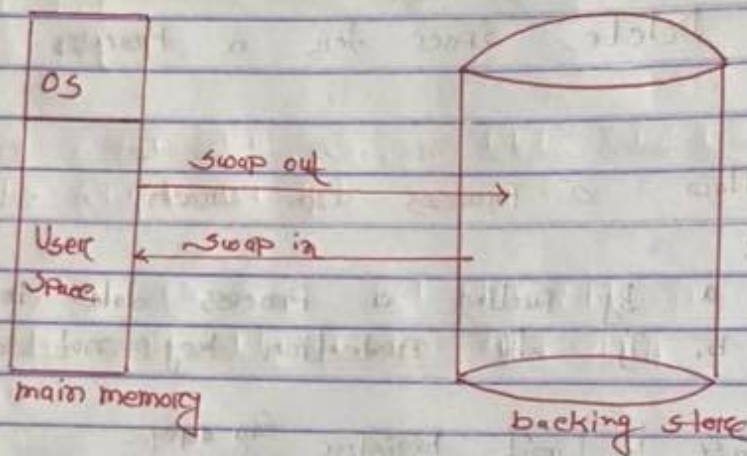
- By putting a Process data into another Process.
- If the Protection key not same.

Base & Limit Register Example -



Base + Location < Limit. If this condition is not satisfied, then the process will crash. If the logical address generated by the CPU is not within the range of the Base & Limit register, then the instruction will not be executed.

Swapping એટલે કેન કોઈ process को Ram से हार्ड डिस्क
 → নিয়ে যাওয়া। Then প্রয়োজনীয় process को
 আবার হार्ড डिस्क থেকে RAM এ নিয়ে আসা।



Swapping થયેલે RAM Է process આવે, અથવા ચા, કોઈ
 process ક્યારેક main memory use કરે ના, ત્યારે hole ડેવી કર,
 એ hole દાનો સમય સંગ્રહ કરેલ Comaction.

Memory usage track કરવા Է બે મુખ્ય techniques

- Bitmaps
- Linked lists.

Q2. What is a bitmap.

Ans:

A representation in which each item corresponds to one or more bits of information.

Bitmap এর সুবিধা অসুবিধা

Bitmap এর সুবিধা ↓

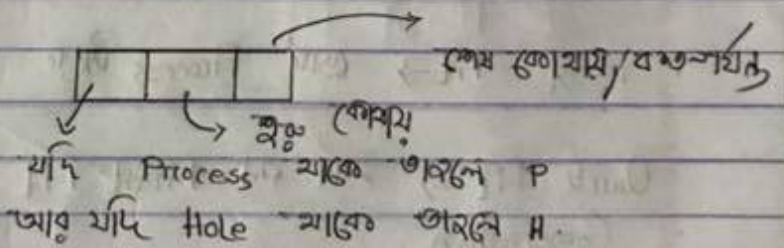
১. অল্পে memory এর check রাখা যায়,

২.

Bitmap এর অসুবিধা ↓

১. সবসময় Zero কোথায় আছে সেটা খুঁজে দেয়া বহুত সময়, ফলে সময় বেশি লাগে,

Bitmap এ জায়গা স্থানি আছে কিনাটার জন্য ০ লিখা হয়, লিখার নিমিত্ত এ space দেখানোর জন্য লিখা হয়.



* Bitmap নিজে থেকে স্থানি জায়গাগুলো মনে রাখবে

* Memory এর কোথায় জায়গা স্থানি সেটা খুঁজে দেয়া বহুত সময় জন্য সুবিধাজনক বন Linked List.

Hole Fill Up এর জন্য চারটি Algorithm follow করা হয়,

- First Fit
- Next Fit
- Best Fit
- Worst Fit
- Quick fit

First Fit $\rightarrow$ শুরু থেকে Search করবে, যেখানে জায়গা খালি আছে, (অথবা বাকি থাকে)

Next Fit $\rightarrow$ শুরু থেকে Search করবে, কখনো প্রথম Process.
 $\rightarrow$ এই প্রথম Process তা বাকি সব থেকে খালি জায়গা Search করবে,

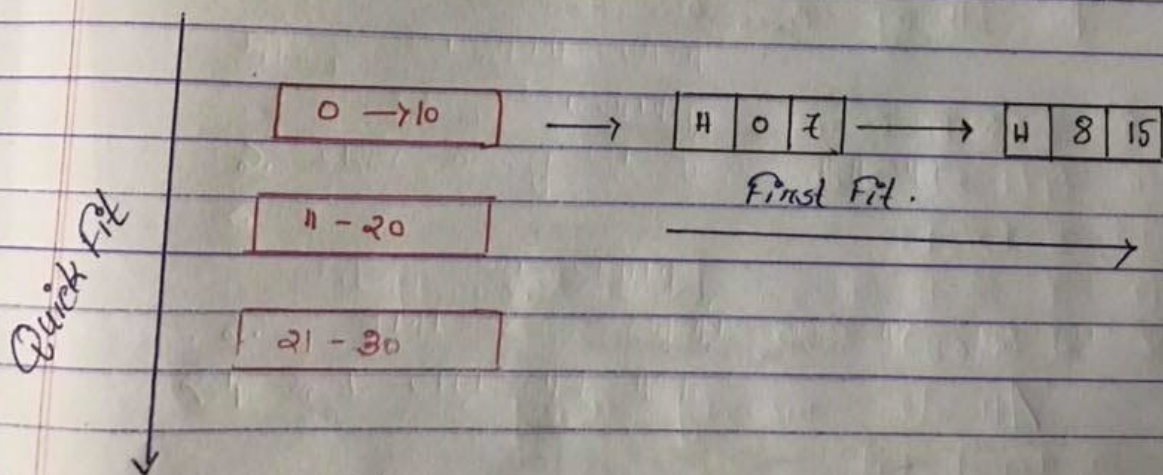
Best fit $\rightarrow$ সব থেকে বাকি জায়গা যেখানে মেথানে ফিট হয়,

Worst fit $\rightarrow$ বড় Process আছে বড় জায়গায় বাকি দেয়,

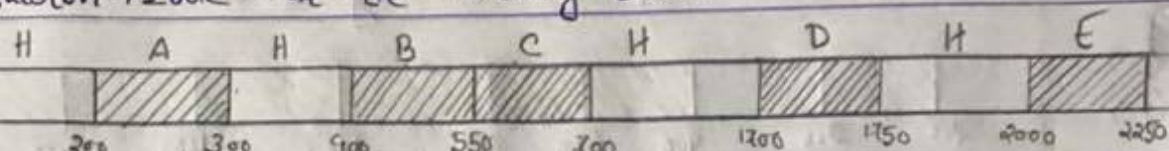
Quick fit $\rightarrow$ বাকি হল First Fit + Quick Fit এর Combination.

অনেকগুলো Memory ফ্রি-স্পেসে ভাগ,

এবং যেকোনো ফ্রি-স্পেসের জন্য Quick Fit. এবং পাশাপাশি
 ফ্রি-স্পেসের জন্য বড়তম ব্লক বসায়, First Fit অথবা অন্য Algorithm
 দ্বারা



Question $\rightarrow$ Look at the memory block.

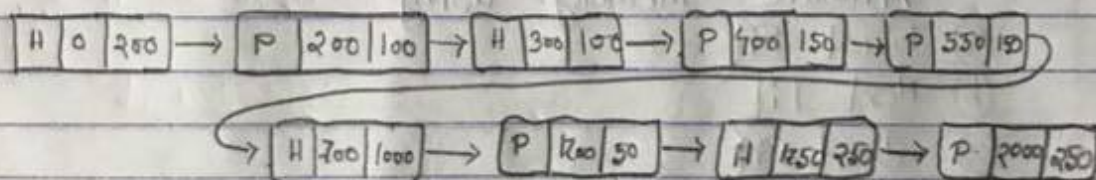


Assuming Per Block 50 bytes. Draw bitmap representation & Linked List representation.

Answer: Bitmap Representation

0000 || 00 || || 00 00 00 00 00 00 00 00 00 00 | 00000 ||||

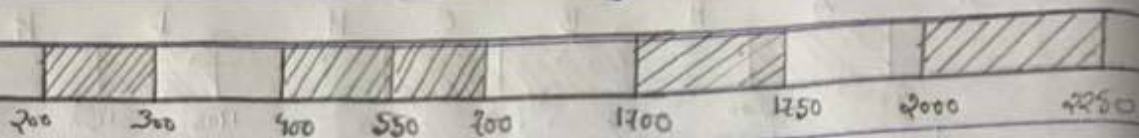
Linked list Representation



কতগুলো ০ আছে আছে মোট ১২টি

(টাকার বি - প্রদত্ত বি) / byte অংশ = বাক্য/লা ০ বা 1
দেখানো হবে,

Question. Look at the memory block



The following cases occurred

- (i) F (220B) Comes
- (ii) G (100B) Comes
- (iii) B Exists
- (iv) J (800B) Comes [Swap Required]
- (v) I (200B) Comes

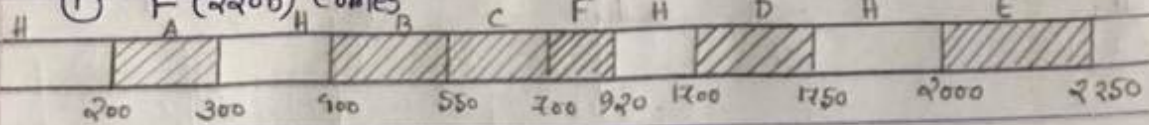
Allocate memory using

- (a) First Fit
- (b) Next Fit
- (c) Best Fit
- (d) Worst Fit

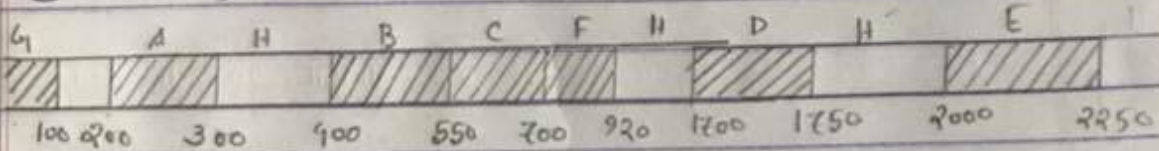
Solve:

(a) First fit

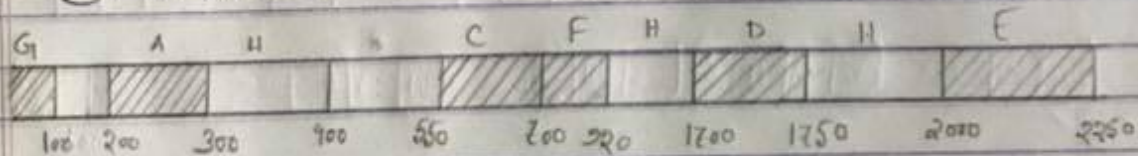
(i) F (2200B) Comes



(ii) G₁ (100B) Comes

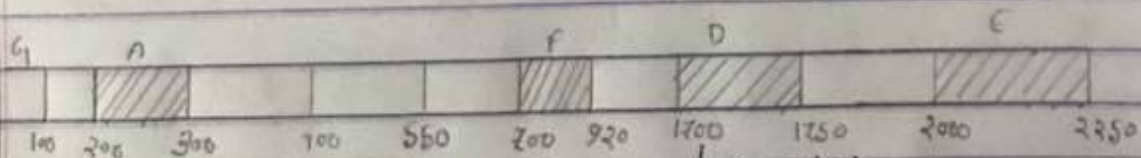


(iii) B Exists



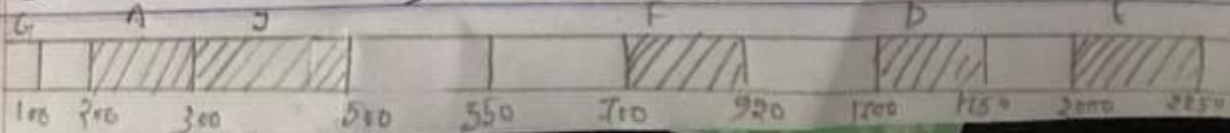
(iv) I (800B) Comes [Swap Required]

Swap করা হলো এর থেকে শুধু প্রকল্প আলাদা করে স্থানি হবে জায়গা।
 কোন পাঠ স্থানি হবে প্রকল্প বলা থাকবে



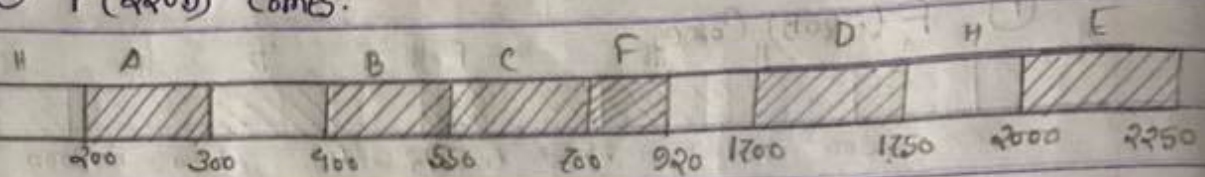
Can not allocate (800B) I.

(v) J (2000B) Comes

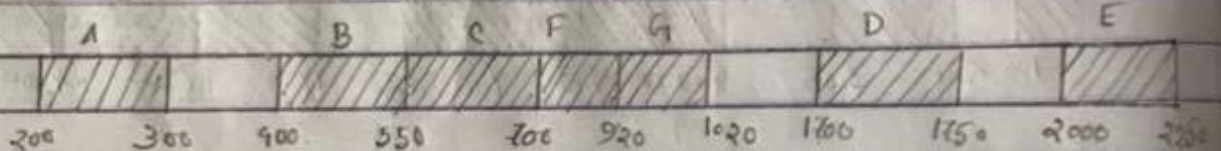


(b) Next fit

(i) F (220B) Comes.



(ii) G (100B) Comes



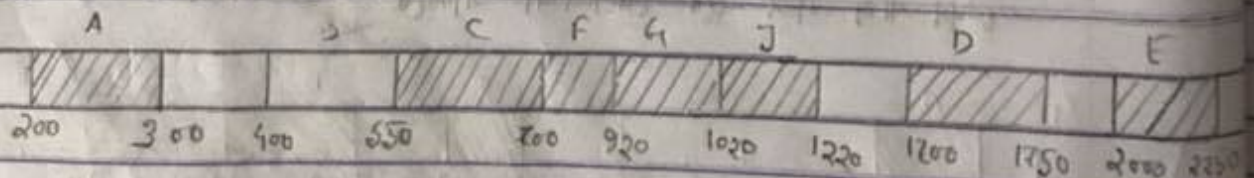
(iii) B exists



(iv) I (800B) Comes

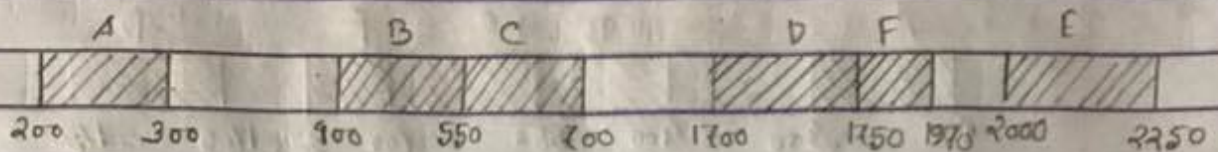
Won't be able to allocate.

(v) J (200B) Comes

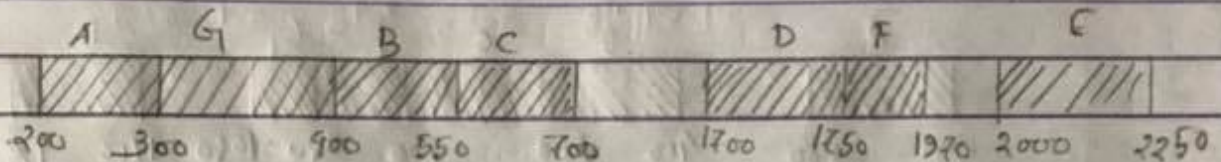


(c) Best fit

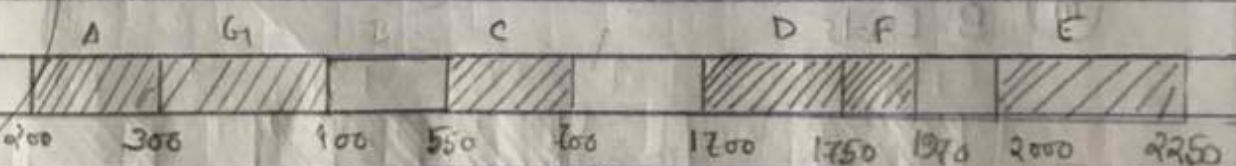
(i) F (220B) Comes.



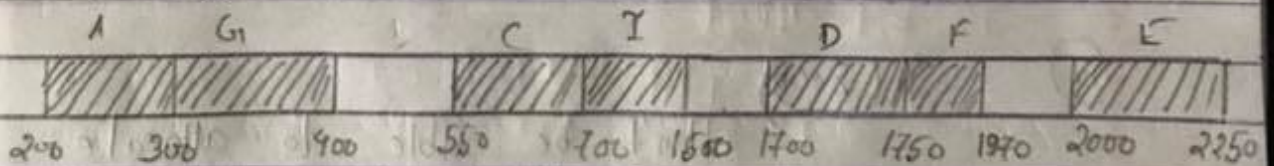
(ii) G₁ (100B) Comes



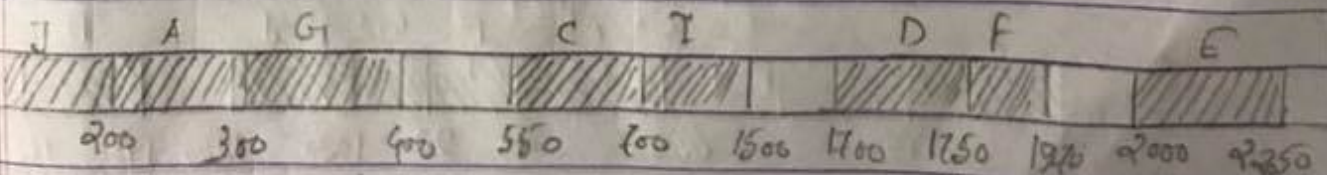
(iii) B exists



(iv) J (800B) Comes

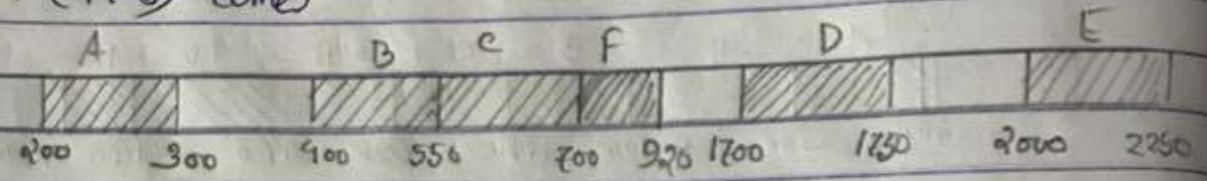


(v) J (200B) Comes.

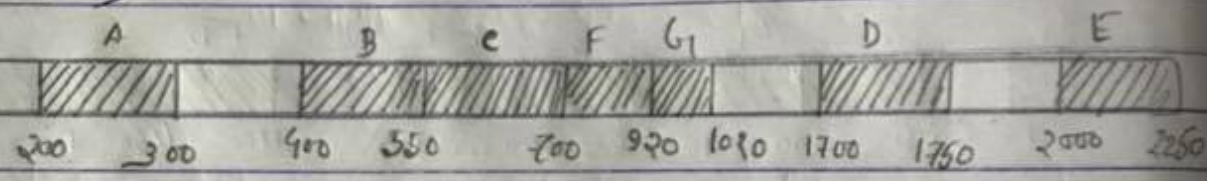


(d) Worst fit

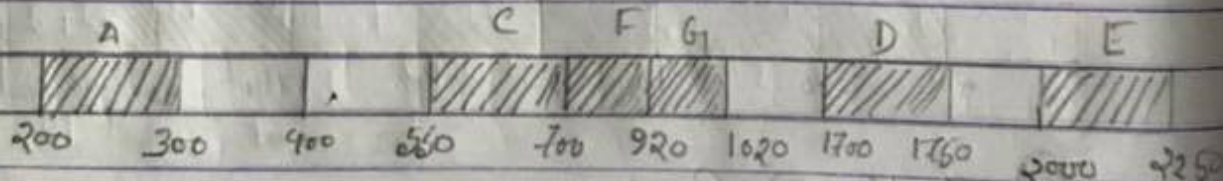
(i) F (200B) Comes



(ii) G₁ (100B) Come



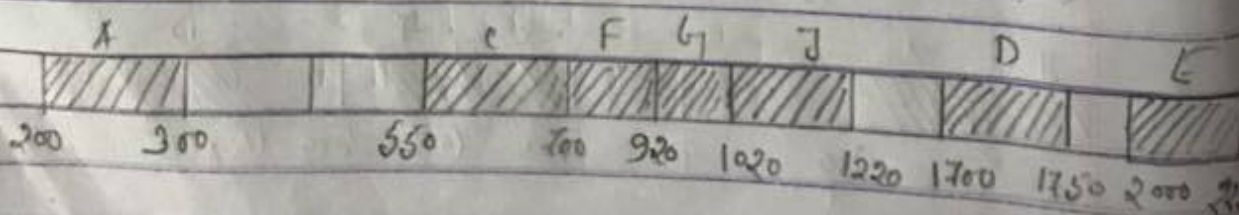
(iii) B Exists



(iv) J (800B)

Won't be able to allocate.

(v) J (200B) Comes



Define Virtual Memory.

Ans: The term virtual memory refers to something which appears to be present but actually it is not.

43. Define Paging.

Ans: Paging is a method of writing data to & reading it from secondary storage for use in Primary storage.

44. Show memory management unit diagram.

Ans:

